MGU-BCA (HONOURS) Regular EXAMINATION – October 2025

Third SEMESTER BCA – Skill Enhancement Course (SEC)

Course Code: MG3SECBCA200 Course Title: PYTHON PROGRAMMING

Time: 1.5 Hours Max Marks: 50

Scheme of Valuation

Part A – Very Short Answer Questions

(Answer all questions. Each question carries 2 marks.) [2 × 10 = 20 Marks]

1. Identify the operator that has higher precedence: and or or. [U]/[CO1]

   Answer: The 'and' operator has higher precedence than 'or' in Python.

2. Write a Python program where a tuple of integers is passed to a function that calculates and returns the sum of all elements. [A]/[CO2]

   Answer:

```
def sum_tuple(t):
    return sum(t)
nums = (10, 20, 30, 40)

s = sum_tuple(nums)
print('Sum =',s)
```

3. Why exception hierarchy is important in Python? [U]/[CO3]

   Answer: Exception hierarchy ensures specific exceptions are caught before general ones, preventing unintended handling.

4. What is the difference between a statement and an expression in Python? [U]/[CO1]

   Answer:

   Statement: performs an action (e.g., assignment).

   Expression: evaluates to a value.

5. Write the syntax of a lambda function. Construct a lambda function to add two numbers. [A]/[CO2]

   Answer:

Syntax: lambda arguments: expression
Example:

```
add = lambda a, b: a + b
print(add(5, 3))
```

6. Discuss the syntax of if…else statement in Python with an example. [K]/[CO1]

   Answer:

```
if condition:
    statement1
else:
    statement2
```

   Example:

```
x = 10
if x > 0:
    print('Positive')
else:
    print('Non-positive')
```

7. Mention the function in math module used to find the square root of a number. [K]/[CO2]

   Answer: math.sqrt(x)
   Example:

```
import math
math.sqrt(16)  # Output: 4.0
```

8. List two libraries in Python used for Data Visualization. [K]/[CO3]

   Answer: Matplotlib, Seaborn

9. Write down the statement to extract the first 4 characters of 'MyPythonWorld'. [U]/[CO1]

   Answer: "MyPythonWorld"[:4] # Output: 'MyPy'

10. State the use of the pickle module in Python. [K]/[CO3]

    Answer: Used for serializing and deserializing Python objects to/from byte streams.

Part B – Short Answer Questions

(Answer any 3 questions. Each question carries 5 marks.) [5 × 3 = 15 Marks]

11. Write a program to print m raise to power n, where m and n are read from the user. [A]/[CO1]

    Answer:

```
m = int(input('Enter base (m): '))
n = int(input('Enter exponent (n): '))

p = m ** n
print(f'{m} raised to {n} is { p }')
```

12. State and explain any five dictionary functions with examples. [K]/[CO2]

    Answer: Explain any 5 dictionary functions with examples

    Functions: keys(), values(), items(), get(), update(), pop(), del, pop(), clear(), popitem(),

13. Write a program that reads a text file and analyzes the frequency of each alphabet letter (a-z). [A]/[CO3]

    Answer:

```
# Open and read the file
filename = ‘example.txt'
with open(filename, 'r') as file:
   text = file.read().lower()  # Convert all text to lowercase
# Initialize dictionary for letters
count = {}
for ch in text:
   if ch.isalpha():  # Count only letters
      if ch in count:
         count[ch] += 1
      else:
         count[ch] = 1
# Print letter counts
for letter in sorted(count):
   print(letter, ":", count[letter])
```

14. Develop a Python program to create a dictionary from two lists and update 'age' to 22. [A]/[CO2]

    Answer:

```
keys = ['id','name','age']
values = [101,'Anu',21]
# Create empty dictionary
```

```
my_dict = {}
# Add key-value pairs using a loop
for i in range(len(keys)):
    my_dict[keys[i]] = values[i]
print("Original Dictionary:", my_dict)

# Update 'age' to 22
my_dict['age'] = 22
print("Updated Dictionary:", my_dict)
```

Output

Original Dictionary: {'id': 101, 'name': 'Anu', 'age': 21}
Updated Dictionary: {'id': 101, 'name': 'Anu', 'age': 22}

15. Write a program to check whether a string is palindrome or not. [A]/[CO1]

Answer:

```
s = input ('Enter string: ')
if s == s[::-1]:
    print('Palindrome')
else:
    print('Not Palindrome')
```

Part C – Essay Questions

(Answer any 1 question. Each question carries 15 marks.) [15 × 1 = 15 Marks]

16. Describe the history of Python in detail. [U]/[CO1]

The answer may include history of Python, features of python or applications etc.

17. Design a mini calculator program using a user-defined function that supports addition, subtraction, multiplication, and division. Make use of a default parameter for the operator. Demonstrate different function calls and outputs. [A]/[CO2]

Answer: provide the code with correct syntax

Prepared by

Dr. Jaseena K U, MES College Marampally Mob: 9847080148